

AREQUIPA, Peru

WMO: 847520

Lat: 16.341S

Lon: 71.583W

Elev: 2562

StdP: 74.10

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.1	6.9	-23.0	0.7	11.0	-20.3	0.8	11.8	10.9	11.6	9.0	12.4	2.9	30	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	10.6	24.0	9.1	23.2	8.8	22.9	8.7	14.3	18.5	13.7	17.8	13.3	17.4	5.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.0	12.9	15.1	12.2	12.2	14.6	12.0	12.0	14.4	49.6	18.4	48.0	17.9	46.6	17.2	16.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	7.0	6.3	DB	3.6	25.3	1.4	1.0	2.6	26.0	1.8	26.6	0.9	27.1	-0.1	27.8	(4)
(5)			WB	-3.2	15.3	0.5	0.5	-3.6	15.6	-3.9	15.9	-4.2	16.2	-4.6	16.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.2	15.6	15.4	15.5	15.5	15.1	14.6	14.4	14.9	15.4	15.4	15.4	15.7	(6)	
	DBStd	1.45	1.53	1.63	1.39	1.41	1.41	1.29	1.35	1.31	1.24	1.31	1.39	1.48	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	1132	87	83	87	84	101	113	122	107	87	90	88	84	(9)	
	CDD10.0	1913	172	151	172	166	158	137	137	151	163	168	162	176	(10)	
	CDD18.3	3	1	1	1	0	0	0	0	0	0	0	1	1	(11)	
	CDH23.3	53	2	2	4	7	7	3	2	4	7	4	3	6	(12)	
(13)	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14)	Wind	WSAvg	3.0	2.7	2.6	2.6	2.7	3.1	3.3	3.3	3.2	3.1	3.0	2.8	(14)	
(15) Precipitation	PrecAvg	87	22	31	22	0	0	0	0	2	1	0	1	4	(15)	
	PrecMax	235	119	121	110	6	3	0	0	37	17	3	10	20	(16)	
	PrecMin	13	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	69	26	33	28	1	1	0	0	7	4	1	2	6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.3	23.4	24.0	24.2	24.2	23.8	23.2	24.0	24.1	24.0	23.9	24.1	(19)	
		MCWB	11.3	13.2	11.5	9.9	8.7	7.8	7.1	7.1	8.8	9.2	8.9	9.6	(20)	
	2%	DB	22.1	22.2	22.2	23.0	23.1	22.7	22.2	22.9	23.1	23.0	22.9	23.0	(21)	
		MCWB	11.3	13.3	12.6	9.8	8.5	7.7	6.9	6.9	7.7	8.6	8.5	9.7	(22)	
	5%	DB	21.2	21.0	21.2	22.0	22.1	21.8	21.2	22.0	22.2	22.1	22.0	22.0	(23)	
		MCWB	11.3	12.8	12.3	10.0	8.0	7.3	6.7	6.9	7.6	7.9	8.4	9.6	(24)	
	10%	DB	20.1	19.9	20.1	20.9	21.0	20.8	20.7	21.0	21.2	21.2	21.0	20.9	(25)	
		MCWB	11.4	12.5	12.1	10.3	7.7	6.7	6.5	6.5	7.3	7.6	8.2	9.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	15.0	14.7	13.6	11.7	10.7	9.3	8.5	10.2	10.9	11.5	13.0	(27)	
		MCDB	18.7	19.2	19.5	18.9	17.4	19.6	18.7	20.2	20.5	19.6	18.9	17.6	(28)	
	2%	WB	13.8	14.4	14.1	12.9	10.6	8.8	8.2	7.7	9.3	9.7	10.7	12.1	(29)	
		MCDB	17.6	18.5	18.6	18.2	17.2	18.9	18.8	19.9	19.8	19.3	18.2	17.3	(30)	
	5%	WB	13.3	13.8	13.6	12.3	9.8	8.0	7.4	7.2	8.6	9.0	9.9	11.4	(31)	
		MCDB	17.2	17.7	17.9	17.7	17.6	18.5	18.7	19.6	19.5	18.9	18.1	17.1	(32)	
	10%	WB	12.7	13.4	13.1	11.7	9.2	7.4	6.8	6.7	7.8	8.2	9.4	10.9	(33)	
		MCDB	16.7	17.1	17.4	17.3	17.9	18.3	18.6	19.3	19.1	18.5	17.9	16.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	8.9	9.7	11.3	12.5	13.0	13.0	12.9	12.6	12.2	11.8	10.6	(35)	
		MCDBR	10.8	10.7	11.3	12.5	13.4	14.0	13.7	13.8	13.4	13.1	12.6	11.9	(36)	
	5% WB	MCWBR	4.6	4.3	4.7	5.8	6.7	6.6	6.5	6.7	6.5	6.1	5.9	5.3	(37)	
		MCWBR	7.9	7.8	8.4	9.6	11.2	11.8	11.7	12.1	11.5	10.9	10.5	9.4	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.423	0.412	0.403	0.390	0.321	0.304	0.300	0.337	0.371	0.376	0.396	0.415	(40)	
		taud	2.117	2.184	2.218	2.213	2.376	2.443	2.449	2.306	2.195	2.179	2.111	2.097	(41)	
	Ebn at Noon	Ebn at Noon	923	924	910	885	918	918	933	925	926	947	942	929	(42)	
		Edh at Noon	170	157	147	138	109	99	100	124	148	156	169	173	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	6.62	6.00	6.22	6.15	5.83	5.44	5.57	6.32	7.06	7.70	8.12	7.48	(44)	
	RadStd	RadStd	0.67	0.64	0.51	0.34	0.22	0.20	0.26	0.23	0.30	0.26	0.29	0.41	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	-0.62	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page